

Lecture 7

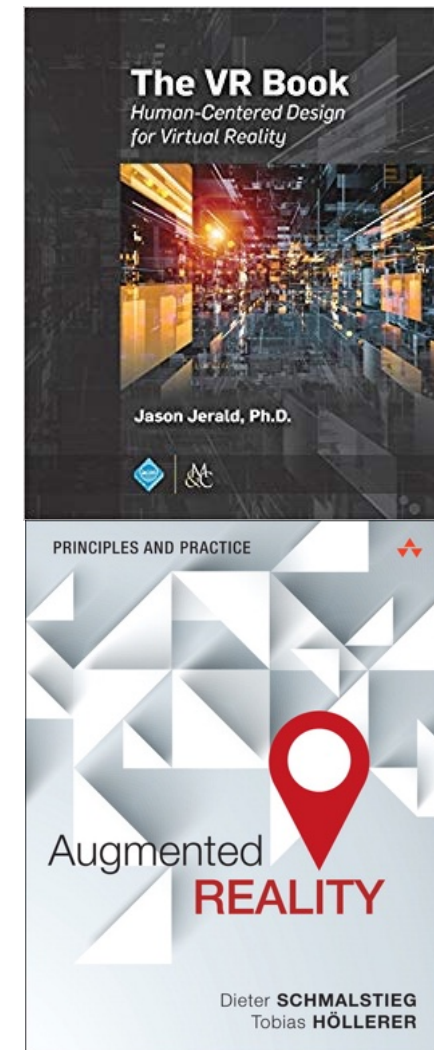
Mixed Reality

Concepts and History



References

- Jason Jerald; **The VR Book: Human-Centered Design for Virtual Reality**, ACM, 2016.
- Dieter Schmalstieg, Tobias Höllerer; **Augmented Reality**, Addison-Wesley, 2016.
- Mark Billinghurst, <https://www.slideshare.net/marknb00/>

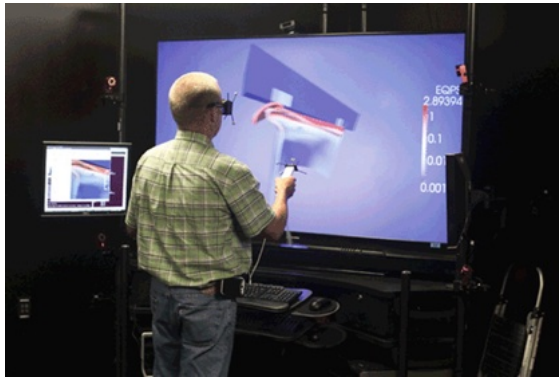


Mixed Reality

- **Mixed Reality Concepts**
- **Virtual Reality**
- **Augmented Reality**
- **History**
- Immersion, Presence and Perception
- MR Technology
- VR Interaction
- Health Issues

Mixed Reality Concepts

Immersive Technologies



JOGOS E AMBIENTES IMERSIVOS - GAMES AND IMMERSIVE ENVIRONMENTS

Fernando Birra
Rui Nóbrega

Milgram's Reality-Virtuality continuum

"...anywhere between the extrema of the *virtuality continuum*."

Mixed Reality

Real
Environment



Augmented
Reality (AR)



Augmented
Virtuality (AV)



Virtual
Environment



Reality - Virtuality (RV) Continuum

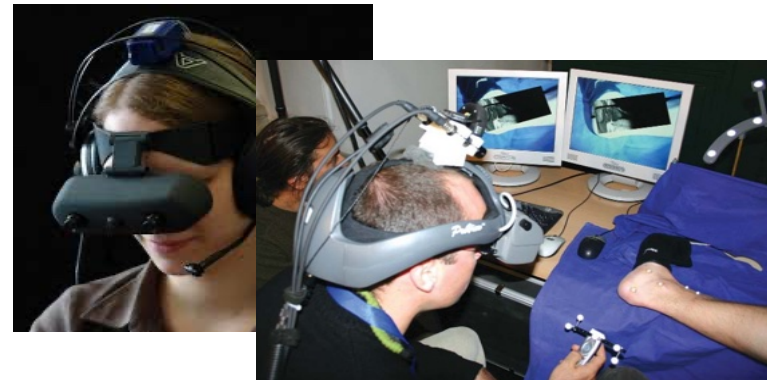
P. Milgram and A. F. Kishino, Taxonomy of Mixed Reality Visual Displays
IEICE Transactions on Information and Systems, E77-D(12), pp. 1321-1329, 1994.

AR/VR Example: The MagicBook



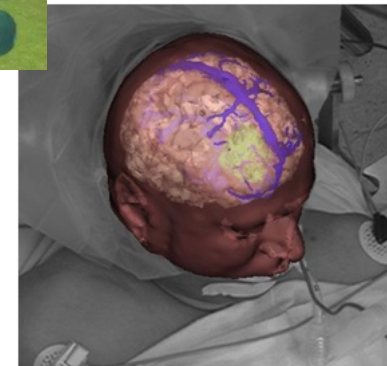
Approaches

- Monitor-based
- Optical see-through
- Video see-through



Applications

- Entertainment / TV
 - Games
 - Virtual Studios/ game annotations
- Medical
 - Guided surgery
- Military
 - Pilot help



Applications

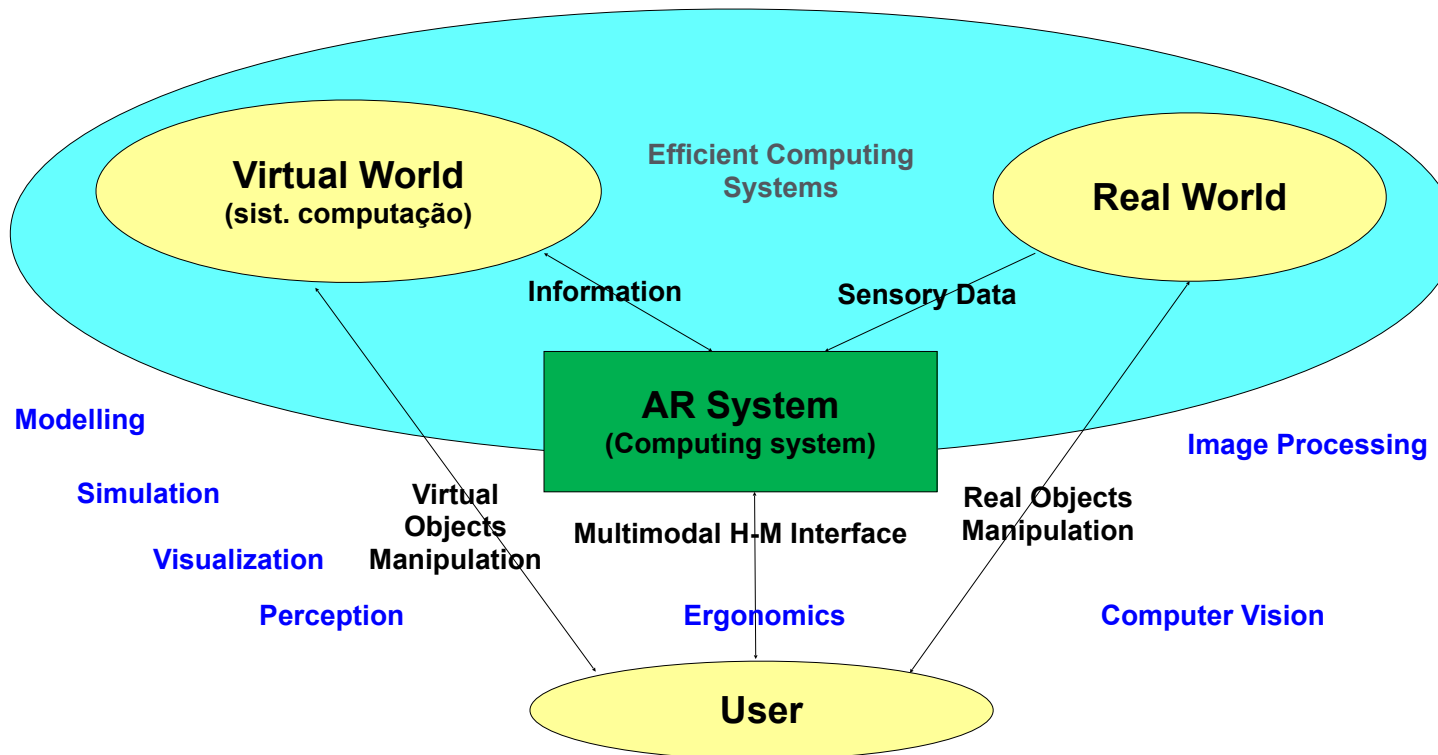
- Engineering, Architecture
 - Collaborative conception
- Maintenance, Repairments
 - Technical diagrams, sequences of maintenance operations, ...
 - Training / equipment training
- Commerce
 - Inserting ads into images
 - Fashion (clothing, hair, ...)
 - Motor vehicle assistance



Technical Challenges

- Virtual Reality
 - user interface
 - detection of user movements
 - latency / side effects
- Augmented Reality
 - alignment between real objects and virtual objects
 - recognition and tracking of reference objects
 - fiducial marks

Connected Areas



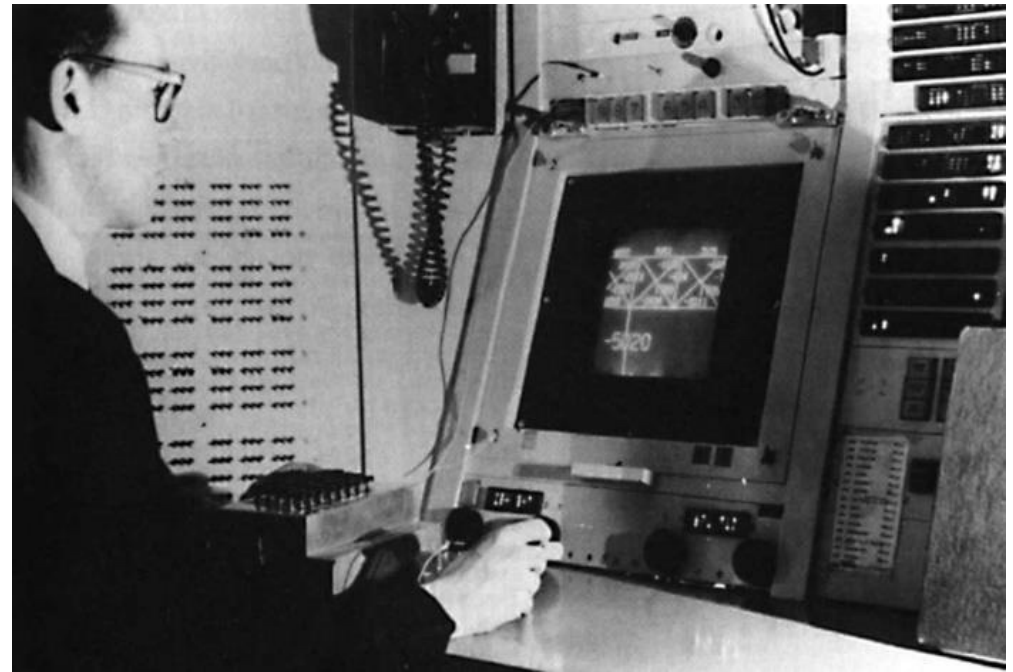
AR links

- <http://computer.howstuffworks.com/augmented-reality.htm>
 - <http://www.augmented-reality-games.com/>
 - <http://www.tomsguide.com/us/augmented-reality-future,news-19099.html>
 - <http://augmentedstories.com/>
 - http://en.wikipedia.org/wiki/List_of_augmented_reality_software
 - <http://www.augmentedreality.org/>
-
- <https://www.youtube.com/watch?v=09vxKN1zLNI>
 - <https://www.youtube.com/watch?v=UWXictuN0wI>

Virtual Reality

Ivan Sutherland (1963)

- Sketchpad – first interactive graphics program



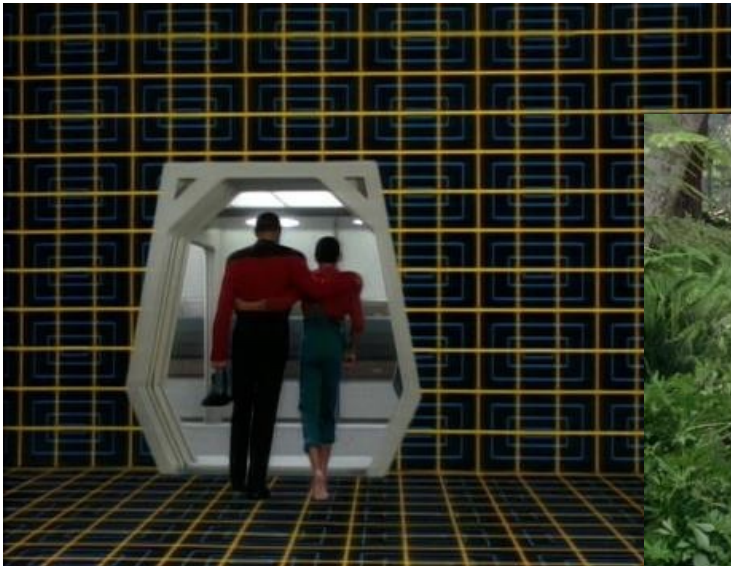
<https://youtu.be/YB3saviltTl>

The Ultimate Display

“The ultimate display would, of course, be a room within which the computer can control the existence of matter. A chair displayed in such a room would be good enough to sit in. Handcuffs displayed in such a room would be confining, and a bullet displayed in such a room would be fatal”.

Ivan Sutherland, 1965

Holodeck (1974)



- First shown in Star Trek; The Animated Series

HoloDeck Video



<https://www.youtube.com/watch?v=oZwtVz7z0wM>



1967 – IBM 1401 – half of the computers in the world, \$10,000/month to run

The Disappearing Computer



1960-70's

Room



1970-80's

Desk



1980-90's

Lap

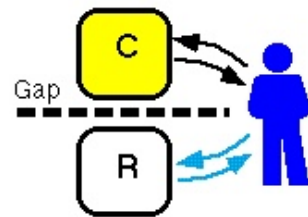


1990-2000's

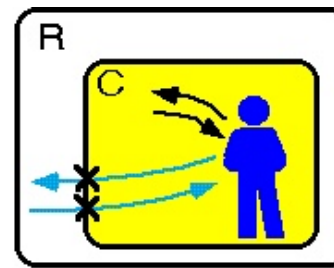
Hand



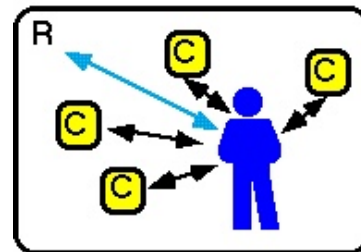
Making Interfaces Invisible



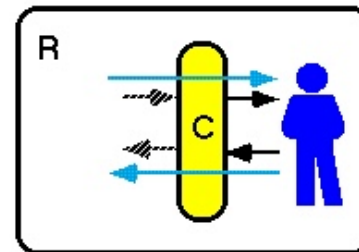
(a) GUI



(b) Virtual Reality



(c) Ubiquitous Computers

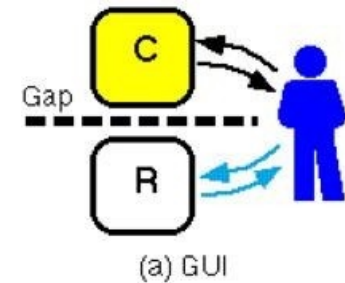


(d) Augmented Interaction

Rekimoto, J. and Nagao, K. 1995. The world through the computer: computer augmented interaction with real world environments. In *Proceedings of the 8th Annual ACM Symposium on User interface and Software Technology. UIST '95*. ACM, New York, NY, 29-36.

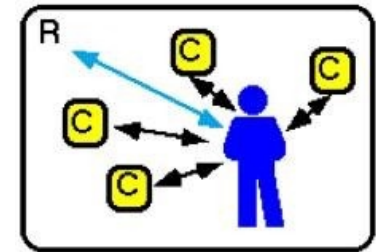
Graphical User Interfaces

- Separation between real and digital worlds
- WIMP (Windows, Icons, Menus, Pointer) metaphor



Ubiquitous Computing/IoT

- Embed computing and sensing in real world
- Smart objects, sensors, etc..

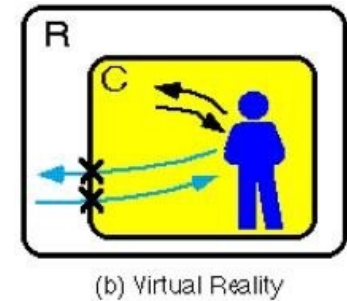


(c) Ubiquitous Computers



Virtual Reality

- Users immersed in Computer Generated environment
- HMD, gloves, 3D graphics, body tracking



What is Virtual Reality?

virtual reality

noun

Simple Definition of VIRTUAL REALITY

Popularity: Bottom 40% of words

: an artificial world that consists of images and sounds created by a computer and that is affected by the actions of a person who is experiencing it

Source: Merriam-Webster's Learner's Dictionary

What is Virtual Reality?

- VR is commonly used by popular media to describe imaginary worlds that only exist in computer and our minds
- **“Virtual Reality is defined to be a computer-generated digital environment that can be experienced and interacted with as if that environment were real”, The VR Book**

Typical VR System



<https://www.youtube.com/watch?v=eJCiyf8Kn9w>

Many Other Definitions

Virtual reality is..

- a computer technology that replicates an environment, real or imagined, and simulates a user's physical presence and environment to allow for user interaction. (Wikipedia)
- electronic simulations of environments experienced via head mounted eye goggles and wired clothing enabling the end user to interact in realistic three-dimensional situations. (Coates, 1992)
- an alternate world filled with computer-generated images that respond to human movements. (Greenbaum, 1992)
- an interactive, immersive experience generated by a computer (Pimentel 1995)

Key Characteristics for VR

Virtual Reality has 3 characteristics:

- 3D stereoscopic display
- Wide field of view display
- Low latency head tracking

When these three things are combined they provide a compelling immersive experience

Defining Characteristics



<https://www.youtube.com/watch?v=FPcbBJbGhmk>

VR Experience



<https://www.youtube.com/watch?v=pAC5SeNH8jw>

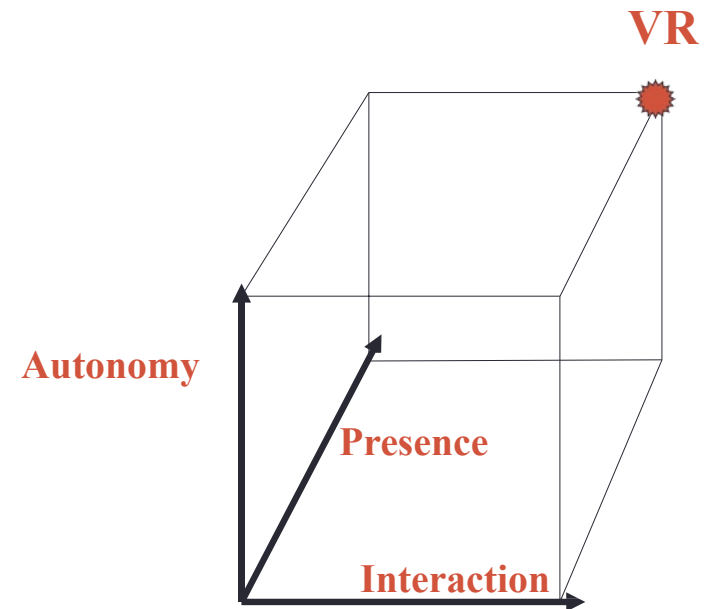
<https://youtu.be/GoSjO9BTBTQ>

Defined in Terms of Presence

- **Presence** is the key to defining VR in terms of experience
- **Presence** is defined as the sense of being in an environment
- **Telepresence** is defined as the experience of presence in an environment by means of a communication medium.
- A “**virtual reality**” is defined as a real or simulated environment in which a perceiver experiences telepresence.

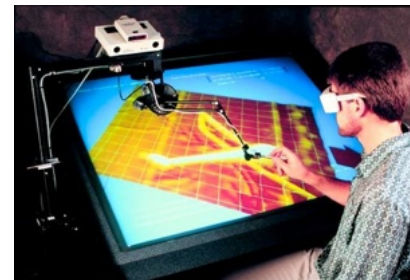
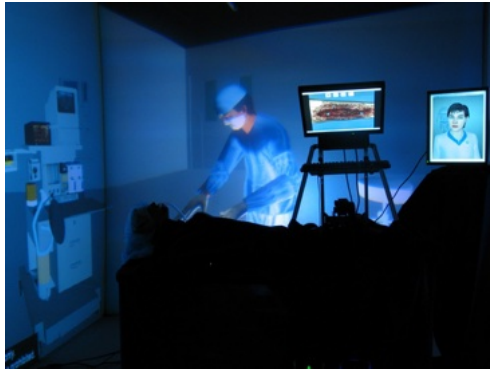
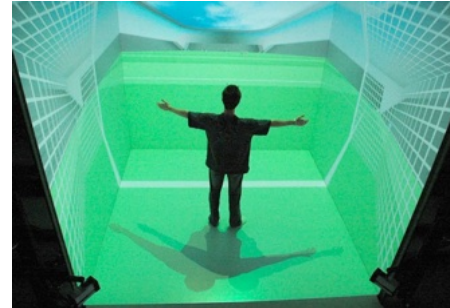
David Zeltzer's AIP Cube

- **Autonomy** – User can react to events and stimuli.
- **Interaction** – User can interact with objects and environment.
- **Presence** – User feels immersed through sensory input and output channels



Zeltzer, D. (1992). Autonomy, interaction, and presence. Presence: Teleoperators & Virtual Environments, 1(1), 127-132.

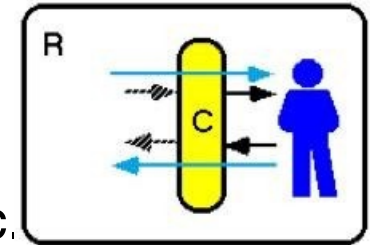
Types of VR



Augmented Reality

Augmented Reality

- Virtual objects blended with the real world
- See-through HMD, handheld display, viewpoint tracking, etc.



(d) Augmented Interaction



Augmented Reality in Science Fiction



1977 – Star Wars

Augmented Reality in Science Fiction



"Who Framed Roger Rabbit?" - Blend real world with computer generated cartoon characters

Augmented Reality Definition

Defining Characteristics [Azuma 97]

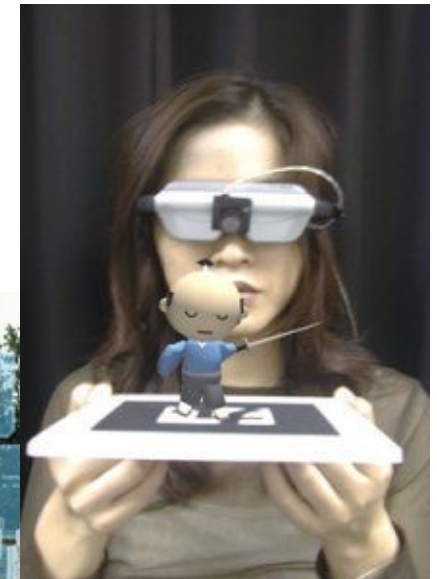
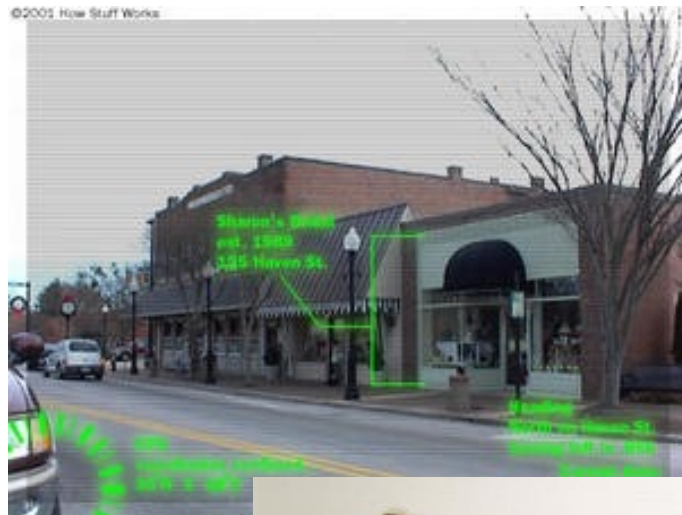
- **Combines Real and Virtual Images**
 - Both can be seen at the same time
- **Interactive in real-time**
 - The virtual content can be interacted with
- **Registered in 3D**
 - Virtual objects appear fixed in space

CNN - 2008



https://www.youtube.com/watch?v=v7fQ_EsMJMs

Augmented Reality Examples



AR Colouring Example - Quiver



<https://www.youtube.com/watch?v=aUPMDwypBkA>

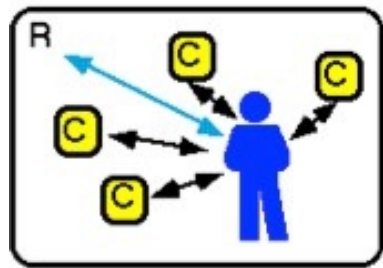
Pokemon Go



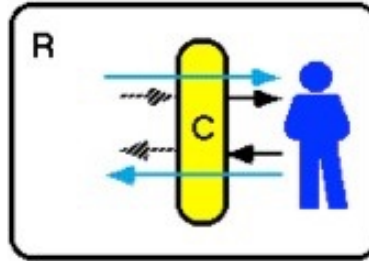
AR vs VR

	Virtual Reality <i>Replaces Reality</i>	Augmented Reality <i>Enhances Reality</i>
Scene Generation	Requires realistic images	Minimal rendering is acceptable
Display Device	Fully immersive, wide field of view	Non-immersive, small field of view
Tracking	Low to medium accuracy is ok	The highest accuracy possible

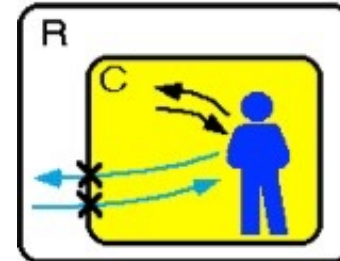
From Reality to Virtual Reality



Ubiquitous Computing



Augmented Reality



Virtual Reality



Real World

Virtual World

Milgram's Reality-Virtuality continuum

"...anywhere between the extrema of the *virtuality continuum*."

Mixed Reality

Real
Environment



Augmented
Reality (AR)



Augmented
Virtuality (AV)



Virtual
Environment



Reality - Virtuality (RV) Continuum

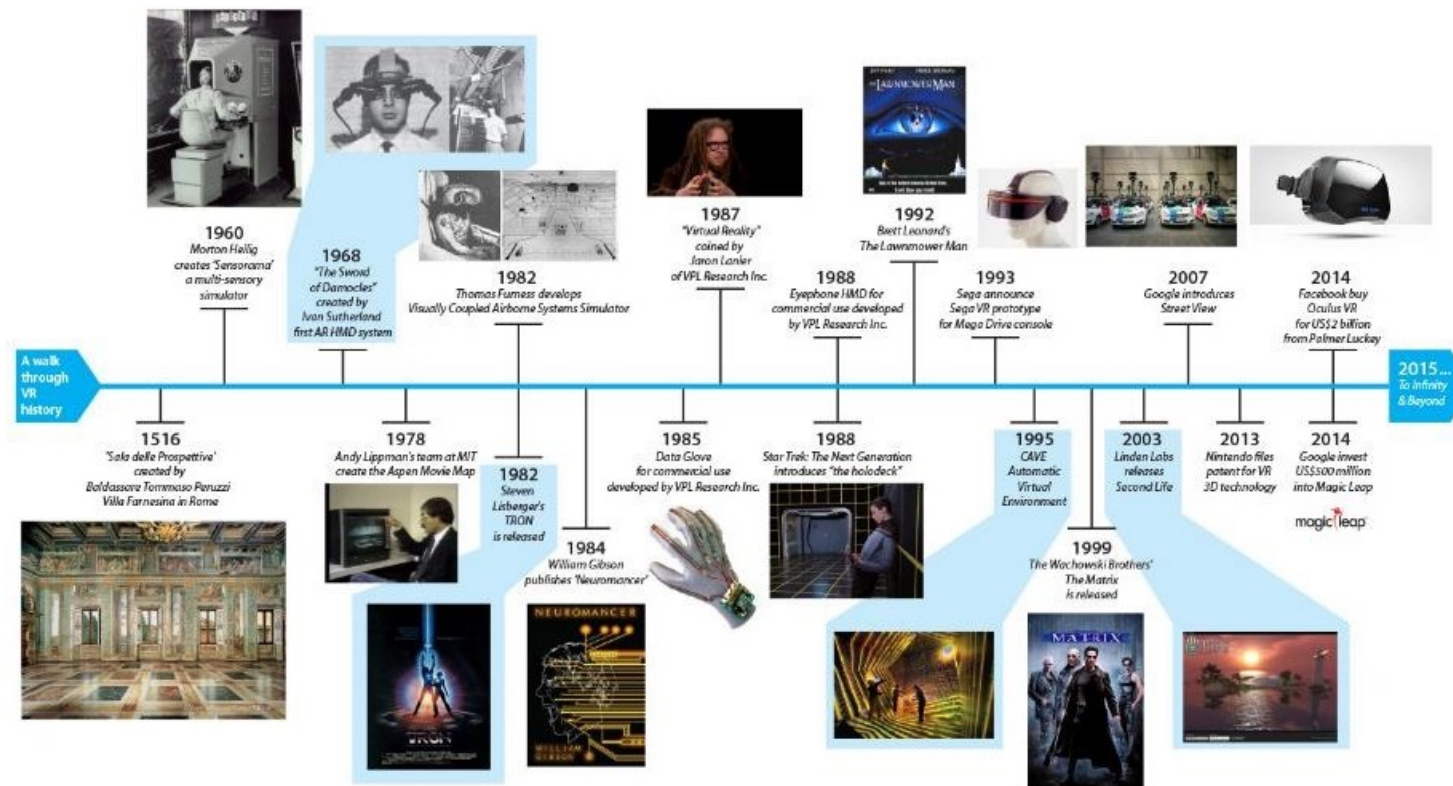
P. Milgram and A. F. Kishino, Taxonomy of Mixed Reality Visual Displays
IEICE Transactions on Information and Systems, E77-D(12), pp. 1321-1329, 1994.

VR Definition Angles

- Virtual Reality can be defined in a number of ways
 - In terms of technology
 - From a Presence perspective
- VR can also be classified with other technologies
 - Invisible Interfaces
 - Milgram's Mixed Reality continuum

History of VR/AR

VR History Timeline



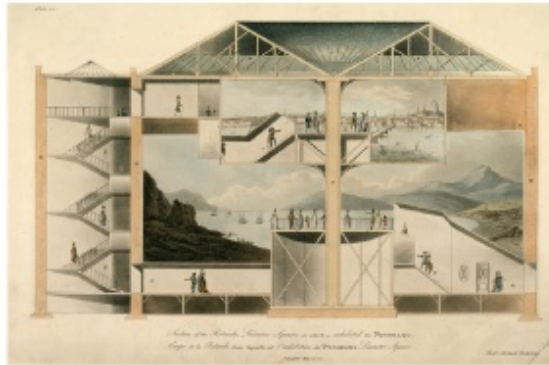
https://immersivelifeblog.files.wordpress.com/2015/04/vr_history.jpg (no longer available)

VR History Timeline

80 BC
Second Style Roman Fresco



1787
Barker's Panorama



1838
Charles Wheatstone's
Stereoscope



1929
Link's Flight Simulator



1962
Heilig's Sensorama



1968
Sutherland's
Sword of Damocles



1939
William Gruber's Viewmaster



1987
Sega Scope
3D glasses





1968
Sutherland's
Sword of Damocles



1987
Sega Scope
3D glasses



1975
Krueger's Videoplace

1985
Jaron Lanier coins the term
"Virtual Reality"



1991
Sega VR
Hype and Flop



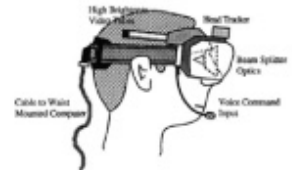
1991
Virtuality Group
Arcade Games



1996
Rekimoto's
NaviCam



1992
Caudell and Minzell
Coin term "Augmented Reality"
as part of Boeing Research Work



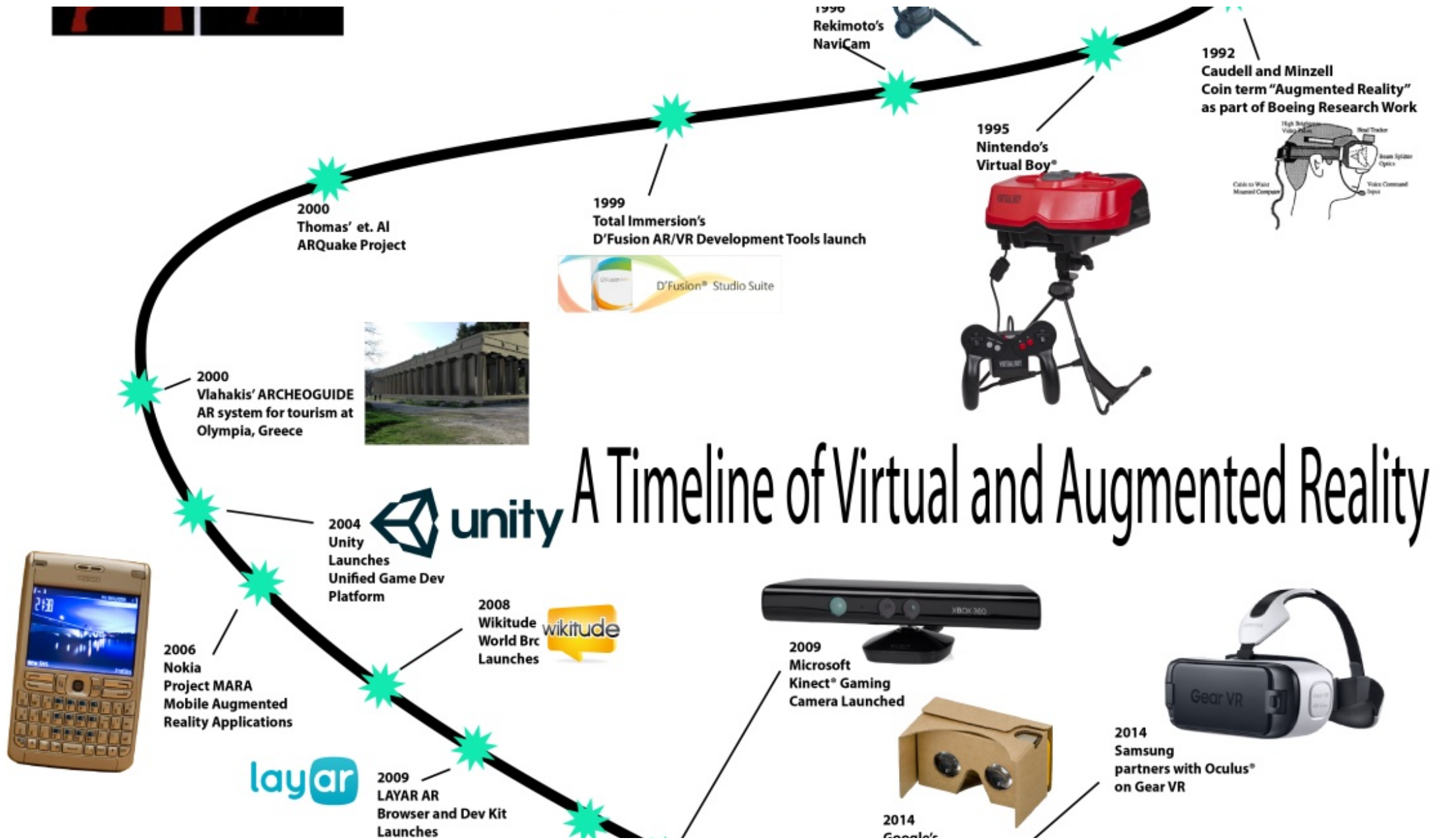
2000
Thomas' et. al
ARQuake Project

1999
Total Immersion's
D'Fusion AR/VR Development Tools launch



1995
Nintendo's
Virtual Boy®





A Timeline of Virtual and Augmented Reality



2006
Nokia
Project MARA
Mobile Augmented
Reality Applications

Unity
Launches
Unified Game Dev
Platform

2008
Wikitude
World Browser
Launches



2009
LAYAR AR
Browser and Dev Kit
Launches



2009
Microsoft
Kinect® Gaming
Camera Launched



2014
Google's
Cardboard
Launches



2014
Samsung
partners with Oculus®
on Gear VR



2012
Palmer Luckey's
Oculus® Launches with
Kickstarter Campaign



2014
Google's
Project Tango
AR Tablet + Phone
Project



2016
Sony
Playstation®VR



2016
Niantic's
Pokémon Go
Launches



2016
Microsoft's
HoloLens®
Launches

2017
Amazon's Lumberyard
Launches VR
Developer Tools



What's
Next?

JOGOS E AMBIENTES IMERSIVOS - GAMES AND IMMERSIVE ENVIRONMENTS

Fernando Birra
Rui Nóbrega

When anything new comes along, everyone, like a child discovering the world thinks that they've invented it, but you scratch a little and you find a caveman scratching on a wall is creating virtual reality in a sense.

Morton Helig (Hammit 1993)

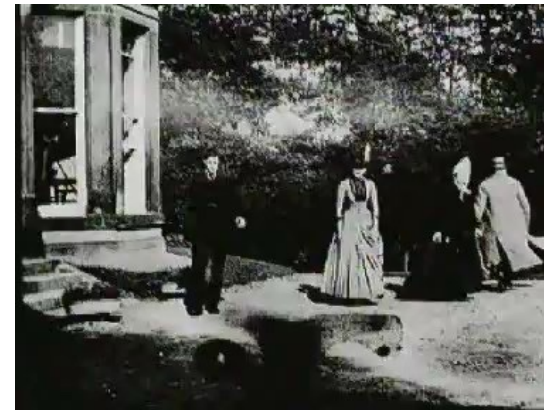
Early History (30,000 BC -)



The history of VR is rooted in human's first attempts to reproduce the world around them

1800's – Capturing Reality

- **Panoramas (1790s)**
 - Immersive paintings
- **Photography (1820-30s)**
 - Oldest surviving photo (Niépce, 1826)
- **Stereo imagery (1830s)**
 - Wheatstone (1832)
 - Brewster (1851)
- **Movies (1870s)**
 - Muybridge (1878)
 - Roundhay Garden Scene (1888)



Stereo Viewers



Wheatstone (1832)



Brewster (1860)

Viewmaster (1939)



3D Cinema 1st Golden Era (1950-60s)



Polarized 3D projection or anaglyph (red/blue)

1900s – Interactive Experiences

- **Early Simulators (<1960s)**

- Flight simulation
- Sensorama (1955)

- **Early HMDs (1960s)**

- Philco, Ivan Sutherland

- **Military + University Research (1970-80s)**

- US Airforce, NASA, MIT, UNC

- **First Commercial Wave (1980-90s)**

- VPL, Virtual i-O, Division, Virtuality
- VR Arcades, Virtual Boy

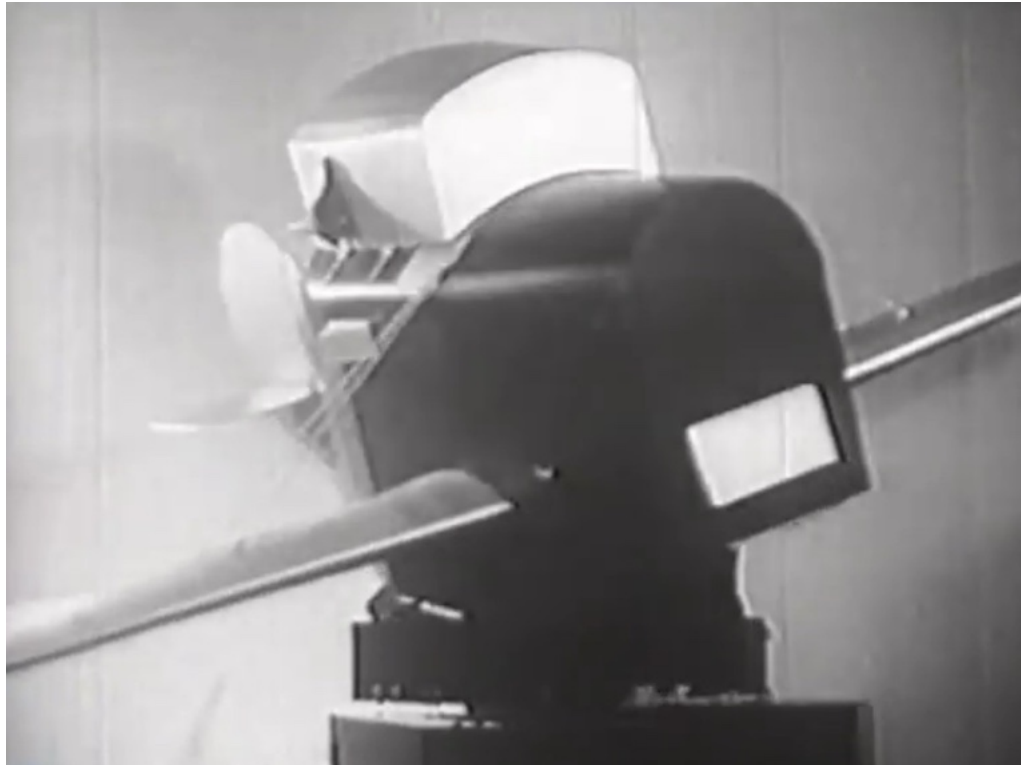


Link Trainer (1929 – 1950s)

- **Flight Simulator Training**
 - Full six degree of freedom rotation
 - Force feedback and motion control
 - Simulated instruments
 - Modeling common flight conditions
- Over 500,000 pilots trained



Link Trainer Video (1940's)



<https://www.youtube.com/watch?v=SjGRLPVq1JU>

Sensorama (1956)

- Created by Morton Heilig
 - Experience Theater
- Multi-sensory
 - Visuals
 - Sound
 - Wind
 - Vibration
 - Smell
- No financial support
- Commercial failure



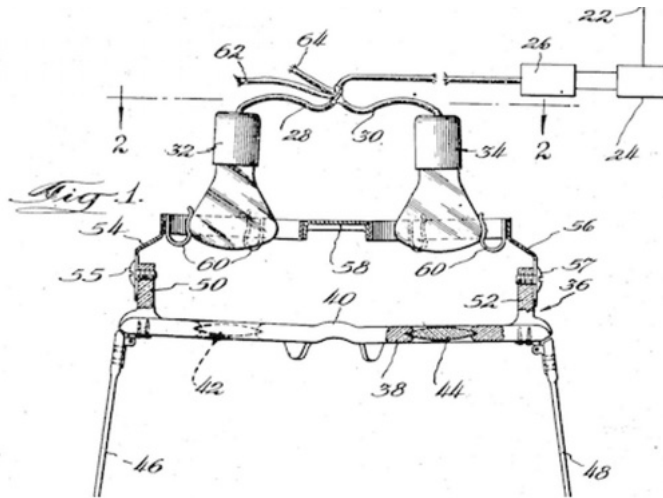
https://www.youtube.com/shorts/N_DDsxgZAwv

Sensorama Video

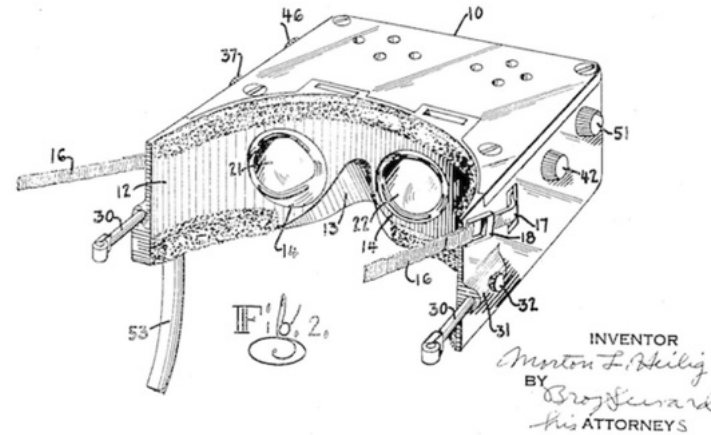


<https://www.youtube.com/watch?v=vSINEBZNCKs>

Early HMD Patents

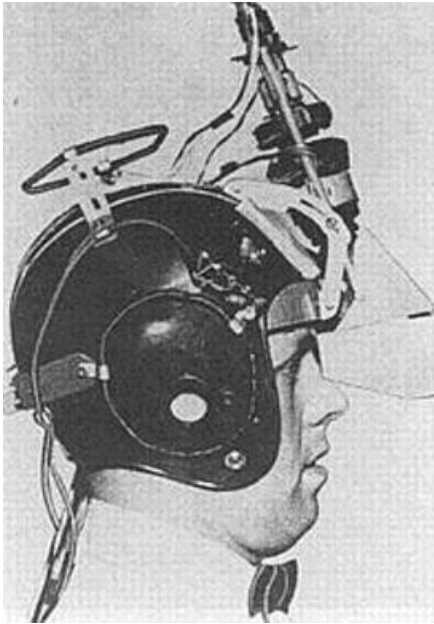


McCollum's Stereo TV HMD (1943)



Heilig's Multisensory HMD (1960)

Early HMDs (1960s)



Philco Headsight (1961)

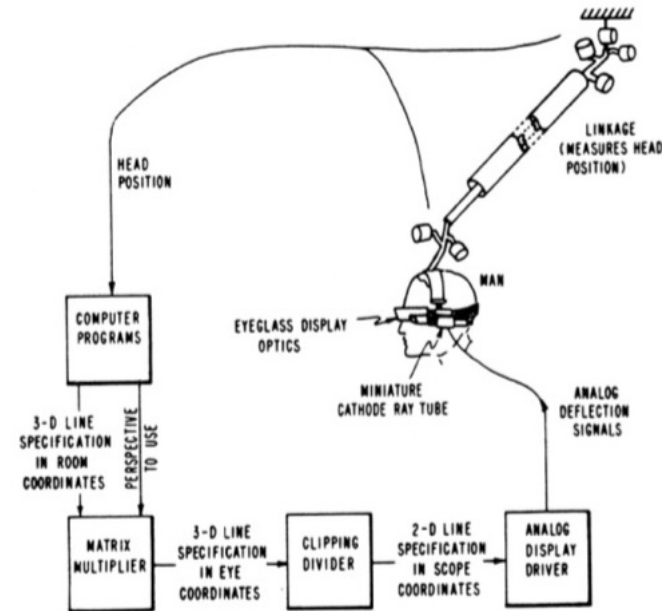
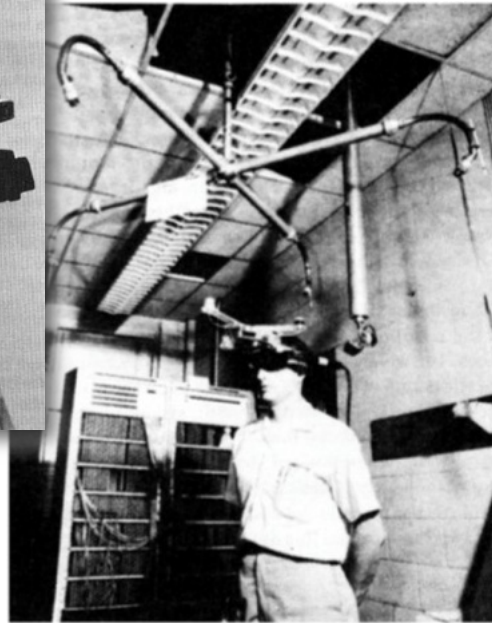


Harvard Viewer (1962)



Sutherland (1968)

Ivan Sutherland (1960s)



Ivan Sutherland's Head-Mounted Display (1968)

Sutherland Display



<https://www.youtube.com/watch?v=NtwZXGprxag>

Super Cockpit (1965-80's)

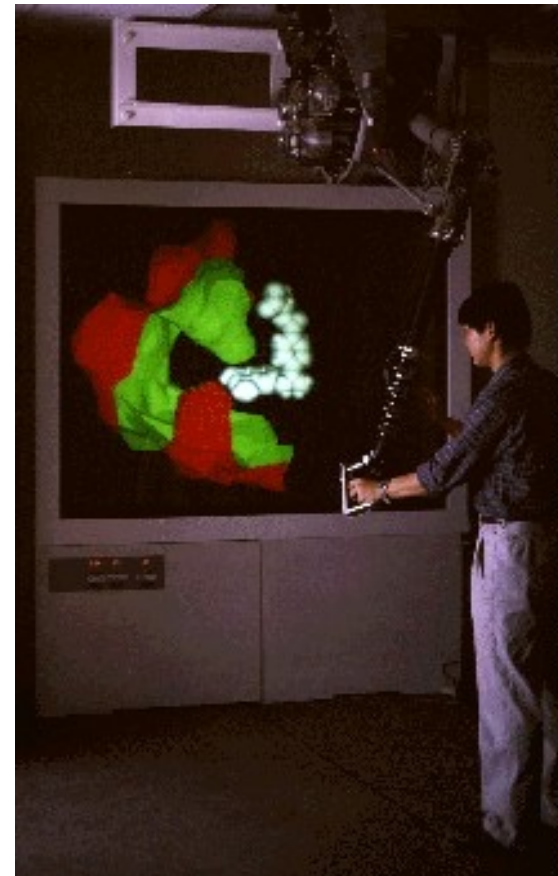
- US Airforce Research Program
 - Wright Patterson Air Force Base
- Tom Furness III
- Multisensory
 - Visual, auditory, tactile
 - Head, eye, speech, and hand input
 - Addressing pilot information overload
 - Flight controls and tasks too complicated
- Research only
 - big system, not safe for ejecting





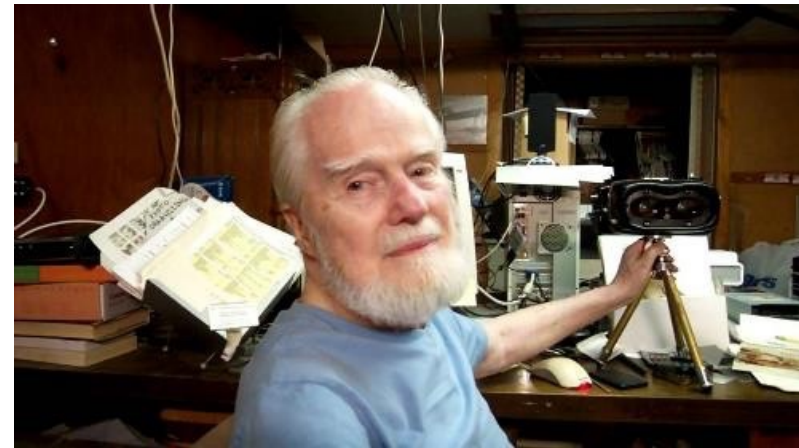
UNC Haptic Systems (1967 – 80's)

- Haptic/kinesthetic display system
 - 6D force fields of molecular structures
- Progression
 - Grope I, simple fields, particle feedback
 - Grope II, 1978, children's building blocks
 - Grope III, late 80's, Remote Manipulator
- Sarcos arm

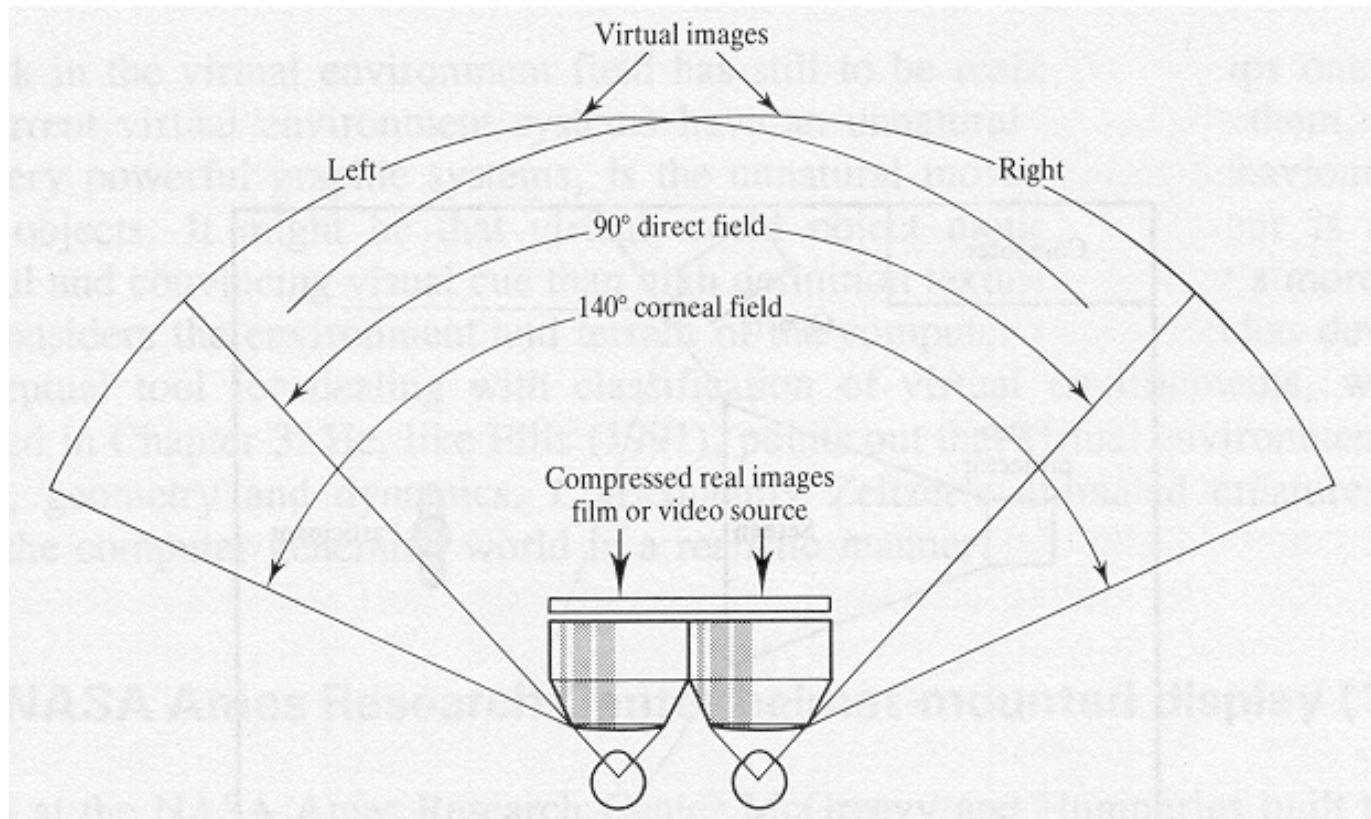


LEEP Optics (1979)

- Large Expanse, Extra Perspective optics
 - Developed by Eric Howlett
- Lens design for extremely wide field of view
 - High resolution in centre, lower resolution in periphery
 - 90° direct FOV, 140° corneal FOV
- Used as basis for most VR HMDs

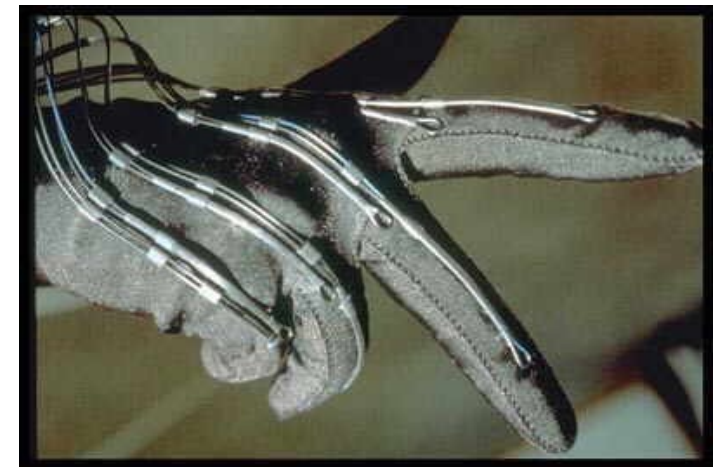


LEEP Optics Design



The Data Glove (1981-82)

- Precursor, Sayre Glove
 - Univ. of Illinois, 1977
 - Thomas Zimmerman (1982)
- Fiber optic bend sensors
 - Detecting finger bending
- Commercialized by VPL
 - Mattel PowerGlove (1989)



VPL DataGlove Demo



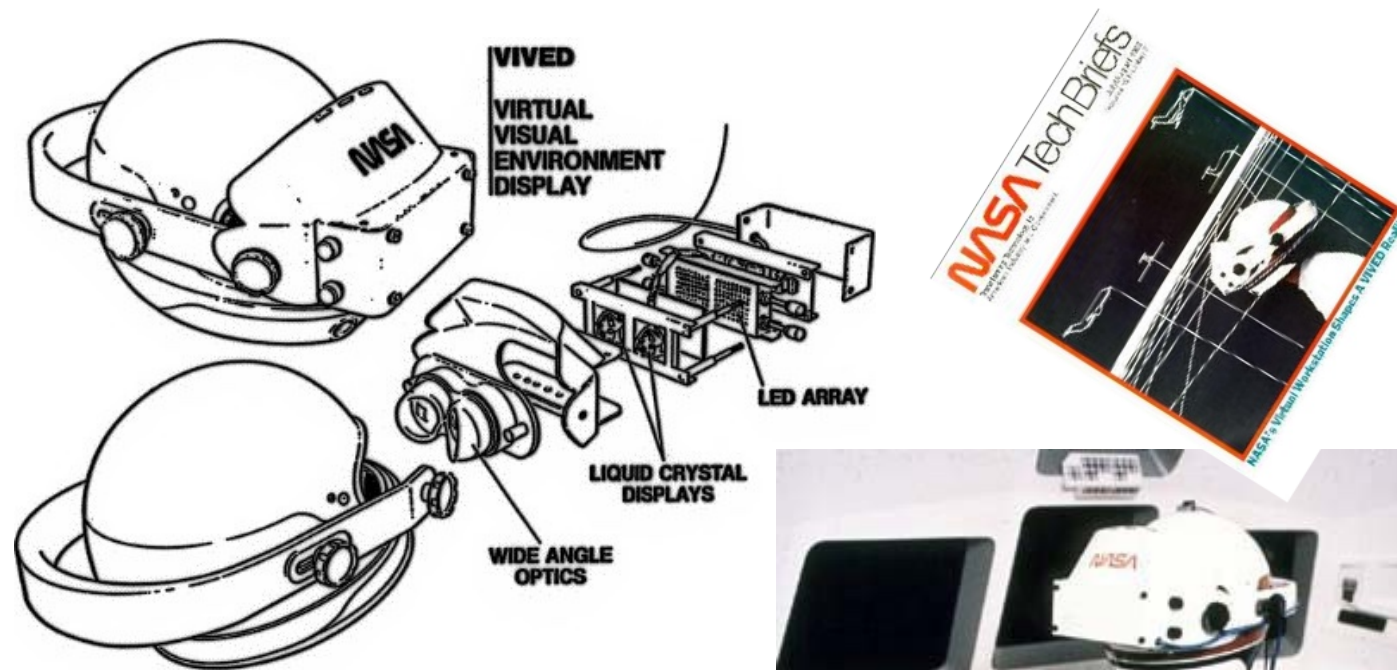
<https://www.youtube.com/watch?v=fs3AhNr5o6o>

NASA VIEW/VIVED (1981-86)

- Early HMD (McGreevy Humphries)
- LCD “Watchman” displays
- VIEW (Scott Fisher)
- Polhemus tracker
- LEEP-based HMD
- 3D audio (Convolvotron)
- DataGlove gesture input
- Simple graphics



Virtual Interface Environment Workstation



Motorbike helmet + LEEP

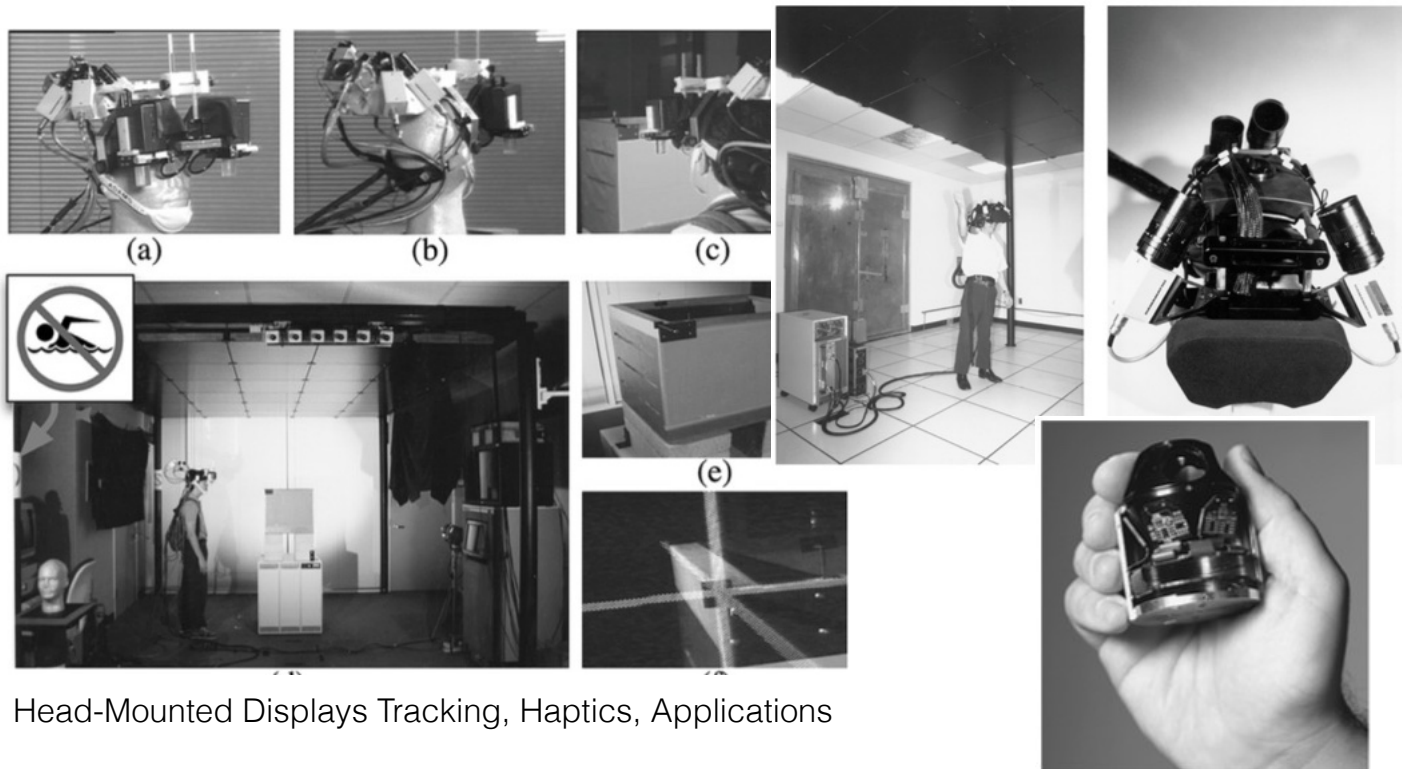


VPL Research (1985 – 1999)

- First Commercial VR Company
 - Jaron Lanier, Jean-Jacques Grimaud
- Provided complete systems
 - Displays, software, gloves, etc
 - DataGlove, EyePhone, AudioSphere

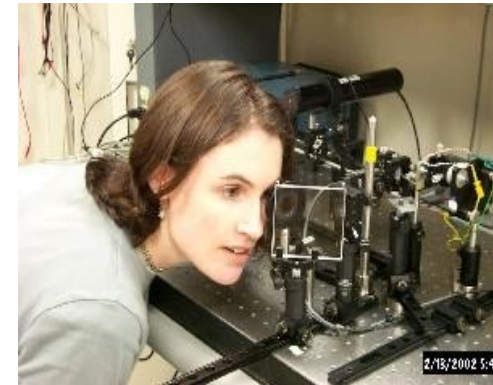


The University of North Carolina at Chapel Hill (1980s-)



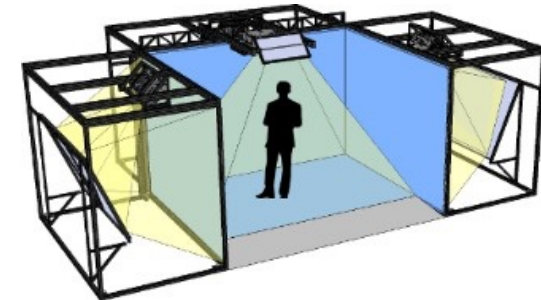
University of Washington (1989 -)

- Human Interface Technology Laboratory (HIT Lab)
 - Founded by Tom Furness III
- Many AR/VR Innovations
 - Virtual Retinal Display
 - ARToolKit AR Tracking library
 - GreenSpace shared VR experience
 - VR and pain care
 - VR and Education



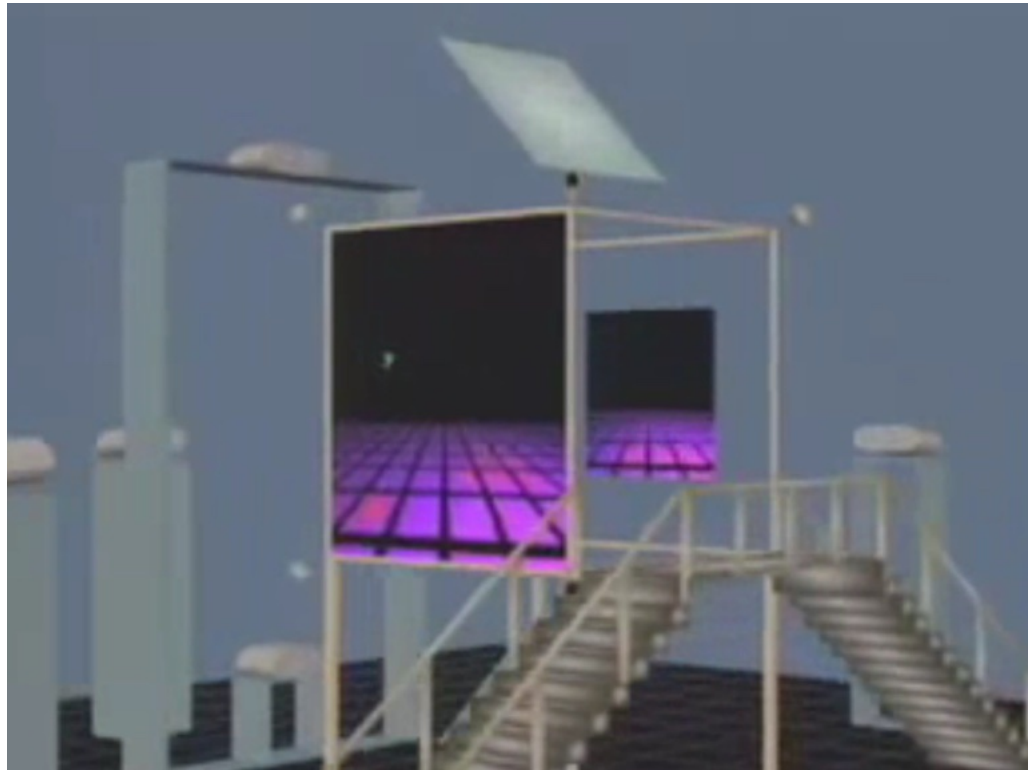
CAVE (1992)

- Projection VR system
 - 3-6 wall stereo projection, viewpoint tracking
 - Developed at EVL, University of Illinois Chicago
- Commercialized by Mechdyne Corporation(1996)



C. Cruz-Neira, D. J. Sandin, T. A. DeFanti, R. V. Kenyon and J. C. Hart. "The CAVE: Audio Visual Experience Automatic Virtual Environment", Communications of the ACM, vol. 35(6), 1992, pp. 64-72.

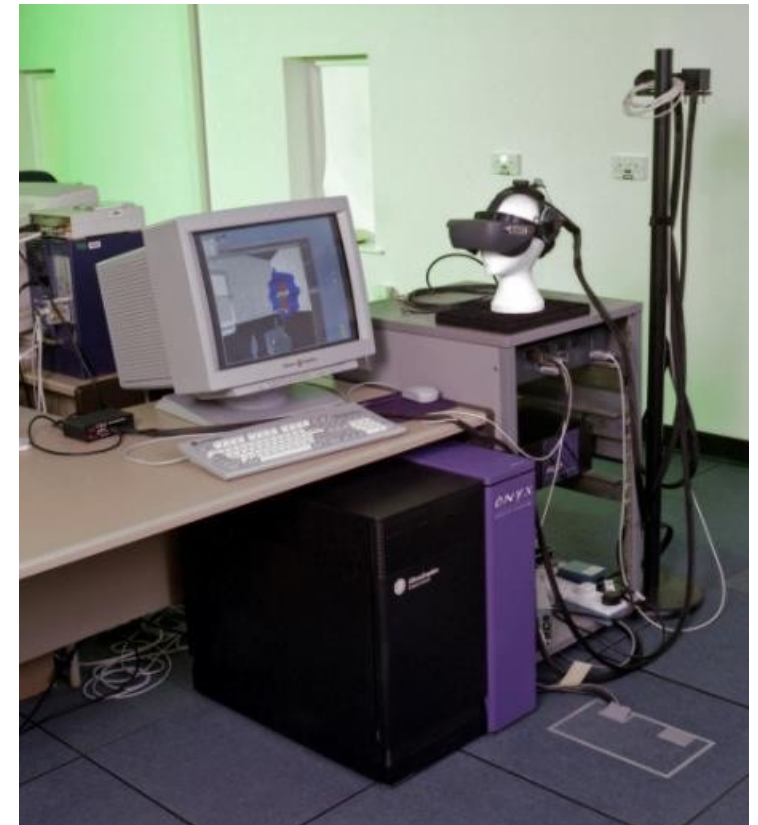
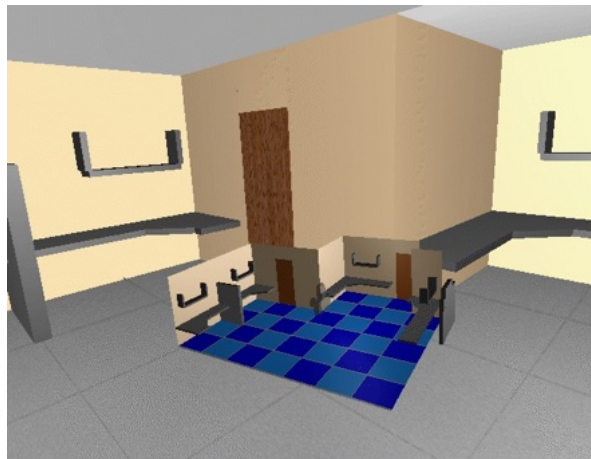
CAVE Demo Video



<https://www.youtube.com/watch?v=aKL0urEdtPU>

Desktop VR - 1995

- Expensive - \$150,000+
- 2 million polys/sec
- VGA HMD – 30 Hz
- Magnetic tracking



Rise of Commercial VR Companies

- Virtual Boy (1995 – 1996)
 - Nintendo's console
- W Industries/Virtuality (1985 - 97)
 - Location based entertainment
- Virtuality VR Arcades
 - Division (1989 – 1998)
- Turn key VR systems
 - Visual programming tools
- Virtual i-O (1993 -1997)
 - Inexpensive gamer HMDs
- Sense8 (1990 - 1998)
 - WorldToolKit, WorldUp
- VR authoring tools



Dactyl Nightmare



https://www.youtube.com/watch?v=hBkP2to1P_c

VR Second Wave (2010 -)

Palmer Luckey

- HMD hacker
- Mixed Reality Lab (MxR) intern



Oculus Rift (2011 -)

- 2012 - \$2.4 million kickstarter
- 2014 - \$2B acquisition FaceBook
- \$350 USD, 110o FOV



The Oculus Kickstarter Video



<https://youtu.be/hxptk8P15TI>

Desktop VR in 2016

Graphics Desktop

- \$1,500 USD
- >4 Billion poly/sec

\$600 HMD

- 1080x1200, 90Hz

Optical tracking

- Room scale



2016 - Rise of Consumer HMDs



Oculus Rift



Sony Morpheus



HTC/Valve Vive

HTC Vive

- Room scale tracking
- Gesture input devices



Example Vive App – Tilt Brush



<https://www.youtube.com/watch?v=ijukZmYFX-0>

VR2GO (2013)

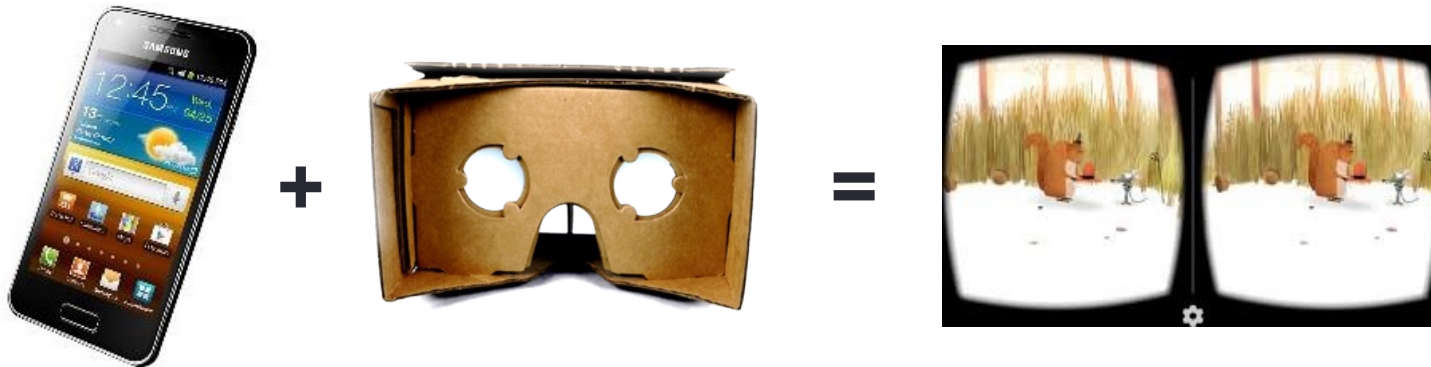
MxR Lab

- 3D print VR viewer for mobiles
- Open source hardware + software
- <http://projects.ict.usc.edu/mxr/diy/vr2go/>
deal link :(
- Early Mobile VR viewer



Google Cardboard

- Released 2014 (Google 20% project)
- >5 million shipped/given away
- Easy to use developer tools



Multiple Mobile VR Viewers Available



Video See-through MR

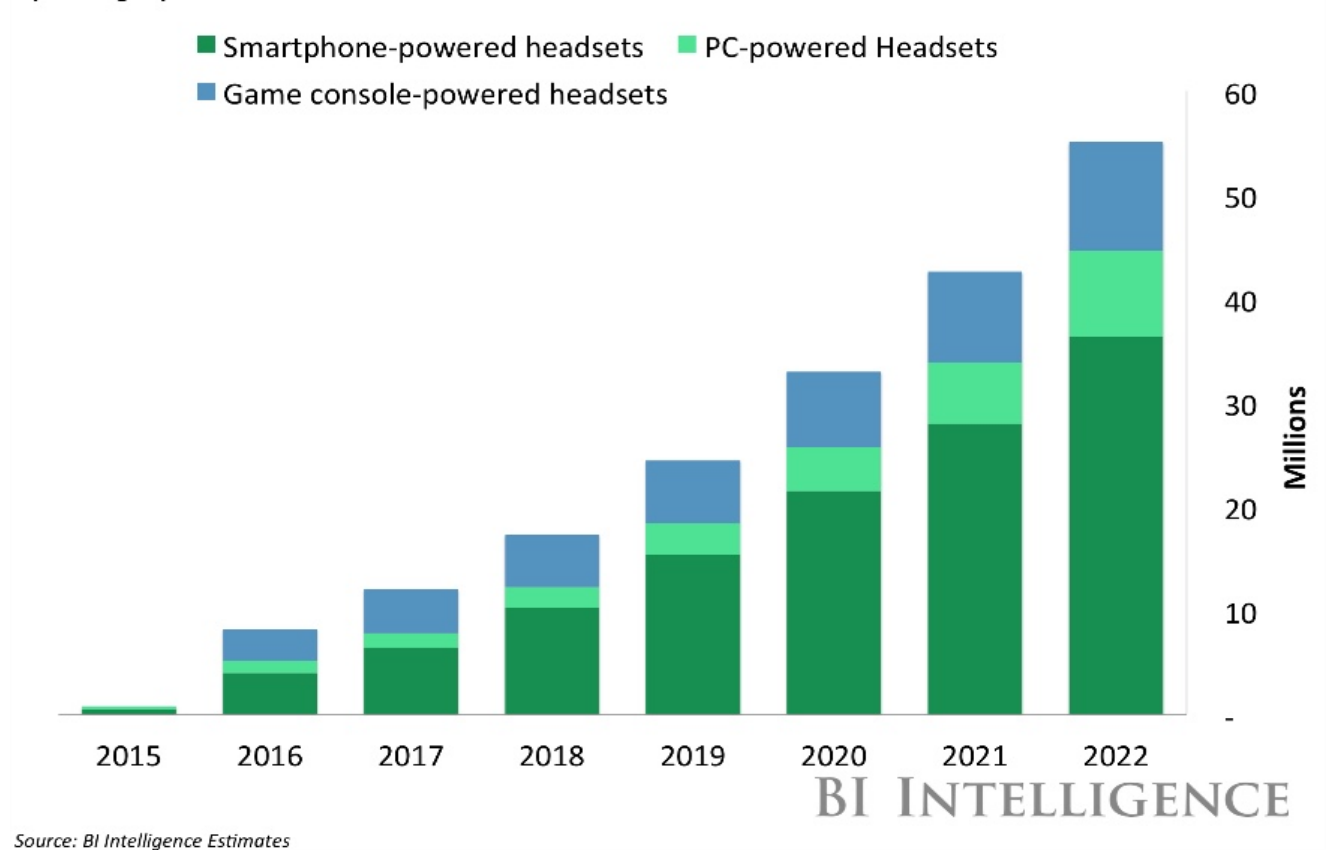
VR headsets with external cameras that capture the world and reproduce it inside VR



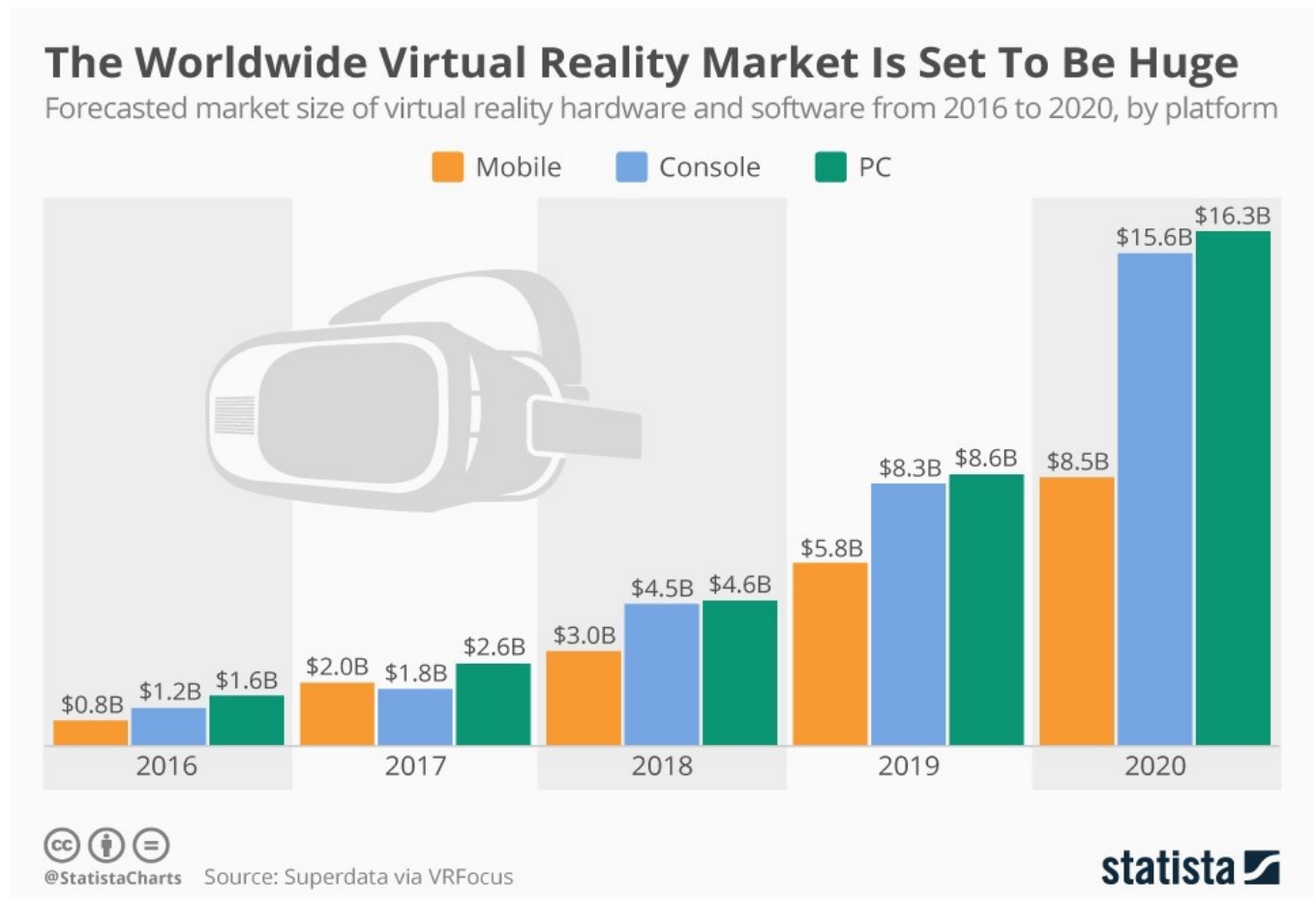
Projected HMD Sales

FORECAST: Global VR Headset Shipments

By Category



Worldwide VR Market Projections



THE VR FUND H2 2017 VR INDUSTRY LANDSCAPE



BY TIPATAT@THEVRFUND.COM

Why 2016 won't be like 1996

- It's not just VR anymore
- Huge amount of investment
- Inexpensive hardware platforms
- Easy to use content creation tools
- New devices for input and output
- Proven use cases – no more Hype! Most important: Focus on User Experience

Conclusion

Virtual Reality has a long history

- > 50 years of HMDs, simulators

Key elements for VR were in place by early 1990's

- Displays, tracking, input, graphics
- Strong support from military, government, universities

First commercial wave failed in late 1990's

- Too expensive, bad user experience, poor technology, etc

We are now in second commercial wave

- Better experience, Affordable hardware
- Large commercial investment, Significant installed user base